

### 1.1.1 Option A – Coastal Landscapes

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| 1. How can coastal landscapes be viewed as systems?                           |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| Key Ideas                                                                     | Content                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1.a. Coastal landscapes can be viewed as systems.                             | <ul style="list-style-type: none"><li>• A conceptual overview of:<ul style="list-style-type: none"><li>◦ the components of coastal landscape systems, including inputs, processes and outputs</li><li>◦ the flows of energy and material through coastal systems</li><li>◦ sediment cells.</li></ul></li></ul>                                                                                                                  |
| 1.b. Coastal landscape systems are influenced by a range of physical factors. | <ul style="list-style-type: none"><li>• Potential influences on coastal landscape systems of:<ul style="list-style-type: none"><li>◦ winds, including speed, direction and frequency</li><li>◦ waves, including wave formation, development and breaking</li><li>◦ tides, including tidal cycles and range</li><li>◦ geology, including lithology and structure</li><li>◦ global pattern of ocean currents.</li></ul></li></ul> |
| 1.c. Coastal sediment is supplied from a variety of sources.                  | <ul style="list-style-type: none"><li>• The various sources of coastal sediment:<ul style="list-style-type: none"><li>◦ terrestrial, including fluvial deposition, weathering and mass movement, marine erosion, aeolian deposition and longshore drift</li><li>◦ offshore, including marine deposition</li><li>◦ human, including beach nourishment.</li></ul></li></ul>                                                       |

## 2. How are coastal landforms developed?

| Key Ideas                                                                                            | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 2.a. Coastal landforms develop due to a variety of interconnected climatic and geomorphic processes. | <ul style="list-style-type: none"><li>• The influence of flows of energy and materials on geomorphic processes, including weathering, mass movement, wave, fluvial and aeolian erosion, transportation and deposition.</li><li>• The formation of distinctive landforms, predominantly influenced by erosion, including bays, headlands, cliffs, shore platforms, geos, blow holes, caves, arches, stacks and stumps.</li><li>• The formation of distinctive landforms, predominantly influenced by deposition, including beaches, spits, on-shore bars, tombolos and salt marshes.</li></ul>                                                                                                                      |
| 2.b. Coastal landforms are inter-related and together make up characteristic landscapes.             | <ul style="list-style-type: none"><li>• <b>Case studies of one</b> high energy coastline (such as rocky) and <b>one</b> low energy coastline, such as estuarine, to illustrate:<ul style="list-style-type: none"><li>◦ the physical factors which influence the formation of landforms within the landscape system</li><li>◦ the inter-relationship of a range of landforms within the characteristic landscape system</li><li>◦ how and why the landscape system changes over time from millennia to seconds, such as cliff collapse in seconds, seasonal changes in beach profile and spit growth over millennia.</li></ul></li></ul> <p>At least <b>one</b> of the case studies must be from beyond the UK.</p> |

## 3. How do coastal landforms evolve over time as climate changes?

| Key Ideas                                                   | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 3.a. Emergent coastal landscapes form as sea level falls.   | <ul style="list-style-type: none"><li>• How landforms in emergent landscapes are influenced by falling sea levels due to a cooling climate, including:<ul style="list-style-type: none"><li>◦ climate changes that occurred during a previous time period and the resultant sea level fall</li><li>◦ the influence of sea level fall and geomorphic processes in shaping landforms, including raised beaches, marine terraces and abandoned cliffs</li><li>◦ the modification of these landforms by processes associated with present and future climate and sea level changes.</li></ul></li></ul> |
| 3.b. Submergent coastal landscapes form as sea level rises. | <ul style="list-style-type: none"><li>• How landforms in submergent landscapes are influenced by rising sea level due to a warming climate, including:<ul style="list-style-type: none"><li>◦ climate changes that occurred during a previous time period and the resultant sea level rise</li><li>◦ the influence of sea level rise and geomorphic processes in shaping landforms, including rias, fjords and shingle beaches</li><li>◦ the modification of these landforms by processes associated with present and future climate and sea level changes.</li></ul></li></ul>                     |

#### 4. How does human activity cause change within coastal landscape systems?

| Key Ideas                                                                                 | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 4.a. Human activity intentionally causes change within coastal landscape systems.         | <ul style="list-style-type: none"><li>• <b>Case study of one</b> coastal landscape that is being managed, including:<ul style="list-style-type: none"><li>◦ the management strategy being implemented and the reason for its implementation, such as groyne construction or off-shore dredging</li><li>◦ their intentional impacts on processes and flows of material, processes and/or energy through the coastal system, such as their effect on the sediment budget</li><li>◦ the effect of these impacts in changing coastal landforms, such as changes in beach profile</li><li>◦ the consequence of these changes on the landscape, such as extension of the coastal landscape seawards.</li></ul></li></ul> |
| 4.b. Economic development unintentionally causes change within coastal landscape systems. | <ul style="list-style-type: none"><li>• <b>Case study of one</b> coastal landscape that is being used by people to illustrate:<ul style="list-style-type: none"><li>◦ the economic development taking place and the reasons for it taking place, such as trade routes, port or tourist resort development</li><li>◦ their unintentional impacts on processes and flows of material, processes and/or energy through the coastal system, such as disturbance to the sediment cell balance</li><li>◦ the effect of these impacts in changing coastal landforms, such as beach profiles</li><li>◦ the consequence of these changes on the landscape, such as coastal retreat or protection.</li></ul></li></ul>       |

#### 5. Topic-specific skills:

- observation skills
- measurement and geo-spatial mapping skills
- data manipulation and statistical skills applied to field measurements
- sediment budget calculations
- mass balance calculations.